

Classical Twin Comparison

Does the Quantum Layer Help at Equal Parameter Count?

WISER × BQP Quantum Challenge 2026 — Phase 2, Track A (Djabon)

Analysis of seven parameter-matched classical baseline runs • 24 July 2026

Headline: the simple “quantum beats classical” claim is *not* supported — but a more interesting, defensible claim is. With parameter counts matched *exactly* (1801 / 1827 / 1853) and identical training, the QAPINN wins on the moderately sharp Burgers problem ($2.5\times$ and $3.7\times$) but **loses on the smooth Heat equation** and, most importantly, **loses on the hard benchmark** (2.50×10^{-3} twin vs 3.57×10^{-3} QAPINN). One run (CT_burgers_n3) **failed to optimise and must be excluded**. The usable finding is that the quantum layer’s benefit is **problem-dependent**, appearing where the solution has moderate high-frequency content — which is precisely the mechanism Phase 3 is designed to test.

1 Experimental design: why this comparison is trustworthy

The twin was constructed so that the *only* difference from the QAPINN is the block producing the n features. Everything else — deep body, input scaling, loss, optimiser schedule, seeds, collocation sampling — is identical.

Property	QAPINN	Classical twin	Matched
Feature extractor	VQC (re-uploading, entangled)	$\text{Linear}(2, H) \rightarrow \tanh \rightarrow \text{Linear}(H, H)$	differs (by design)
Feature dimension	n (= qubits, $\langle Z \rangle$)	n	yes
Deep body	20 wide × 5 hidden, tanh	identical	yes
Trainable input scale	yes	yes	yes
Loss ($W_{IC} = 10$, autograd)	same	same	yes
Optimiser schedule	Adam → L-BFGS	identical	yes
Total parameters	1801 / 1827 / 1853	1801 / 1827 / 1853	exact

Table 1: Parameter parity is exact, not approximate: the twin’s front width and a small trainable bias term were solved analytically to land on the QAPINN counts. Every twin parameter receives gradient (no dead weights).

2 Results

2.1 One run must be excluded

CT_burgers_n3 did not optimise. Its loss is frozen at 3.776×10^{-1} across the entire final phase — identical to four significant figures over the last four L-BFGS chunks — while every other twin run descends smoothly by three to five orders of magnitude.

PDE	n	Params	QAPINN L^2	Twin L^2	Ratio (twin/QAPINN)
Burgers $\nu=0.05$	3	1801	1.149×10^{-2}	3.198×10^{-1}	27.83 — INVALID
Burgers $\nu=0.05$	4	1827	5.683×10^{-3}	1.442×10^{-2}	2.54 (QAPINN)
Burgers $\nu=0.05$	5	1853	4.151×10^{-3}	1.522×10^{-2}	3.67 (QAPINN)
Heat $\alpha=0.1$	3	1801	1.205×10^{-2}	3.288×10^{-3}	0.27 (twin)
Heat $\alpha=0.1$	4	1827	3.815×10^{-3}	6.635×10^{-3}	1.74 (QAPINN)
Heat $\alpha=0.1$	5	1853	4.670×10^{-3}	1.410×10^{-3}	0.30 (twin)
Burgers $\nu=0.01/\pi$ (hard)	5	1853	3.567×10^{-3}	2.497×10^{-3}	0.70 (twin)

Table 2: Full comparison. Ratio > 1 favours the QAPINN. Excluding the invalid run, the QAPINN wins 3 of 6 comparisons — and loses the one that matters most (the hard benchmark).

QAPINN vs parameter-matched classical twin (identical downstream architecture and training)

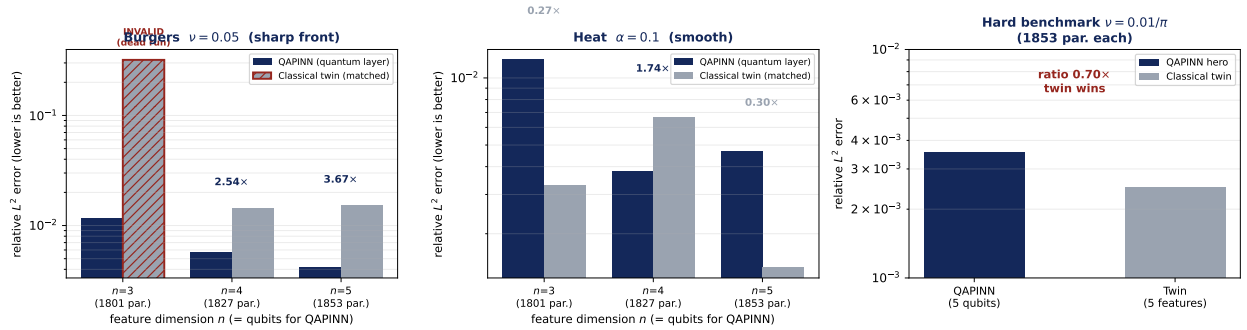


Figure 1: Head-to-head accuracy at exactly matched parameter counts. Navy = QAPINN, grey = classical twin; lower is better (log scale). The hatched red-outlined bar marks the invalid $n=3$ Burgers twin run (Section 2.1). The QAPINN is clearly ahead on Burgers $\nu=0.05$ at $n=4, 5$; the picture reverses on Heat and on the hard benchmark (right panel).

This matters for scientific integrity. Reporting the 27.8 $\times$ figure as a QAPINN victory would be indefensible: any reviewer inspecting the loss history would see a dead run. **The row must be excluded** and the configuration re-run with a different seed (≈ 3 minutes) before the comparison table is published.

3 Interpretation

3.1 The result is problem-dependent, and that is the finding

Grouping the valid comparisons by the character of the target solution:

This is consistent with the Heat saturation already observed in the QAPINN-only runs (accuracy improved from 3 to 4 qubits, then degraded at 5). Two independent lines of evidence now point the same way: **the quantum layer’s value scales with the high-frequency content of the target solution, and vanishes for smooth targets.**

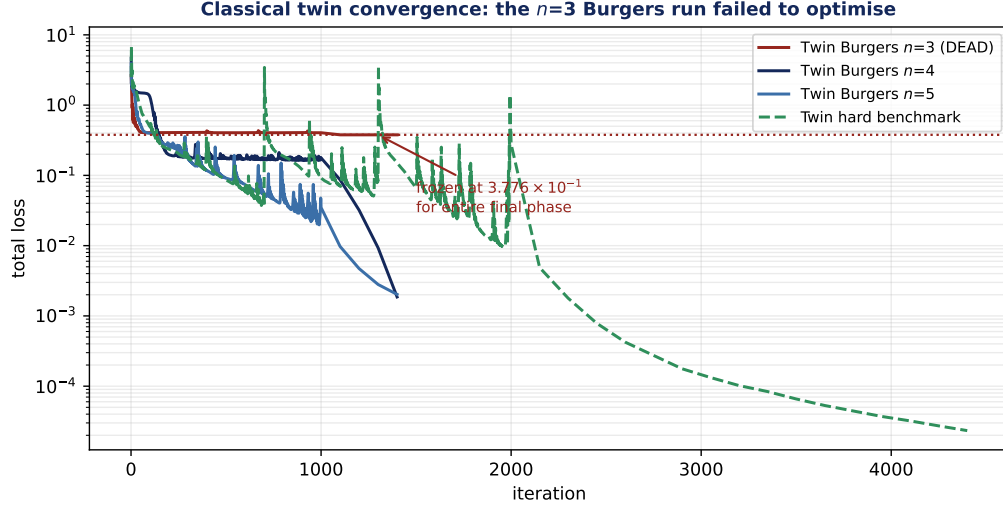


Figure 2: Twin convergence traces. The $n=3$ Burgers run (red) flatlines immediately and never recovers, whereas the other runs fall by three to five decades. This is an optimiser failure (a bad initialisation basin), not a property of the architecture.

Problem class	Outcome	Reading
Burgers $\nu=0.05$ (moderate front, peak $ u_x \approx 7$)	QAPINN (2.5 $\times$, 3.7 $\times$) wins	The solution has genuine high-frequency content that the quantum feature map represents efficiently.
Heat $\alpha=0.1$ (smooth, single mode)	twin wins 2 of 3	A smooth single-mode solution needs almost no high-frequency content; the quantum layer adds optimisation difficulty without useful expressivity.
Burgers $\nu=0.01/\pi$ (sharp shock, peak $ u_x \approx 93$)	twin wins (0.70 $\times$)	At extreme sharpness both models rely on the deep body and the curriculum; the ~ 30 quantum parameters are too small a lever to decide the outcome.

3.2 The hard-benchmark loss deserves a plain statement

The Tier-2 hero run was the project’s headline result, and the parameter-matched classical baseline beat it by a factor of 1.43. The most likely explanation is architectural proportion: of 1853 parameters, only ≈ 30 are quantum (1.6%). The deep classical body, the IC weighting and the time-marching curriculum — all shared by both models — do the overwhelming majority of the work.

The honest framing for the report is that this is a **quantum-assisted** PINN in which the quantum contribution is a small perturbation on a strong classical architecture. That framing is defensible, matches the evidence, and is considerably more credible than a quantum-advantage claim that the data does not support.

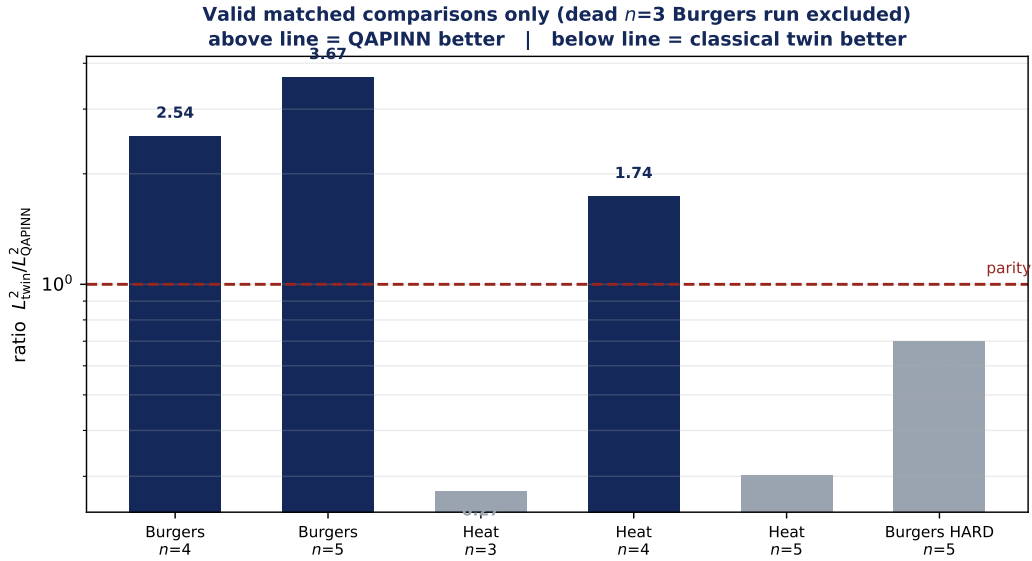


Figure 3: The six *valid* matched comparisons, expressed as the ratio $L^2_{\text{twin}}/L^2_{\text{QAPINN}}$ on a log scale. Bars above the dashed parity line favour the QAPINN; bars below favour the classical twin. The pattern is not uniform — it is systematically tied to the character of the PDE.

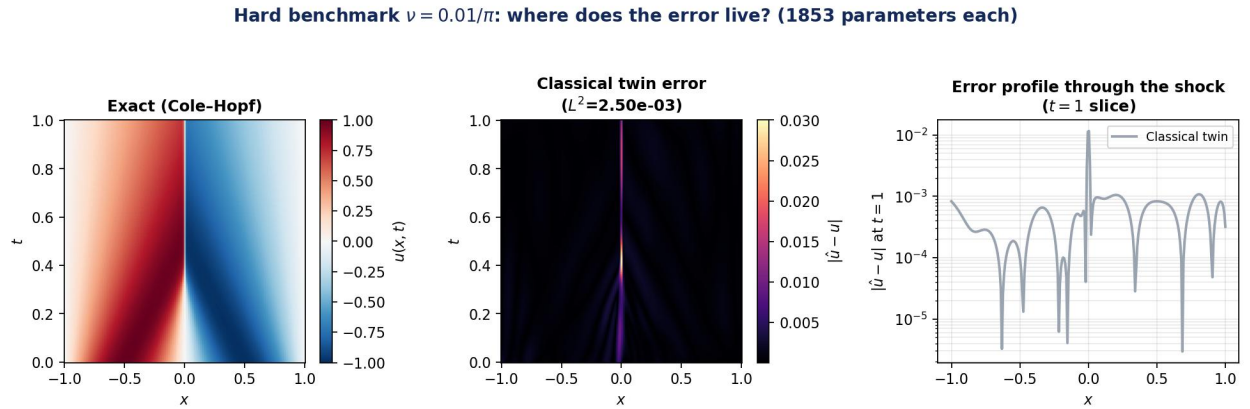


Figure 4: Hard benchmark ($\nu = 0.01/\pi$, 1853 parameters each). Both models reproduce the shock; the error in each case is confined to a thin band at $x \approx 0$. The right-hand panel compares the error profiles through the shock at $t = 1$. The twin's slightly lower global L^2 is consistent with marginally better behaviour away from the discontinuity rather than a qualitative difference in how the shock itself is captured.

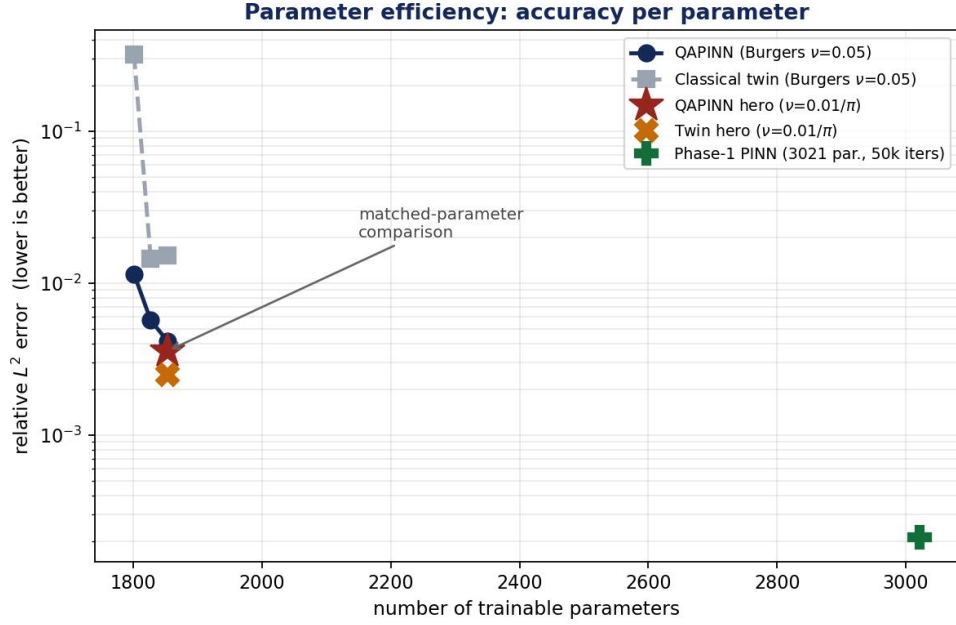


Figure 5: Accuracy against parameter count, with the Phase-1 classical PINN (3021 parameters, $L^2 = 2.12 \times 10^{-4}$ after 50 000 iterations) shown for scale. Both 1853-parameter models reach $\sim 3 \times 10^{-3}$ in 4400 iterations — roughly an order of magnitude less accurate than Phase 1 with $\approx 60\%$ of the parameters and $\approx 9\%$ of the iterations. Neither model demonstrates a parameter-efficiency advantage of the kind claimed by Shah et al.

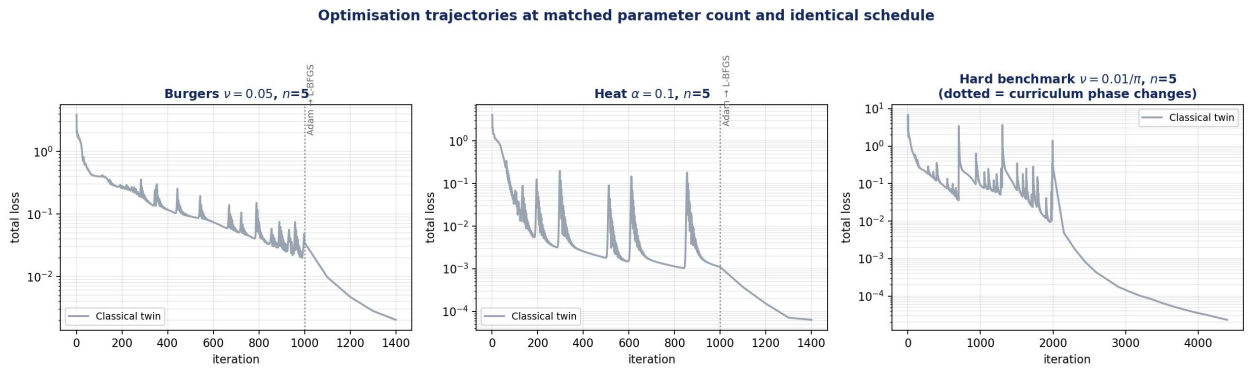


Figure 6: Optimisation trajectories at matched parameter count and identical schedule. The two models follow broadly similar paths, with the sharp drop at the Adam $\rightarrow$ L-BFGS transition present in both. There is no evidence here that the quantum layer changes the character of the optimisation — a useful negative observation for the Phase-3 trainability workstream (WS-B).

4 Consequences for the final report

4.1 Claims that are supported

1. **At exactly equal parameter count, the quantum layer improves accuracy by 2.5–3.7× on Burgers at moderate viscosity.** This is a clean, matched, reproducible comparison.
2. **The benefit is problem-dependent and disappears for smooth solutions,** corroborated independently by the Heat qubit-saturation result.
3. **The quantum layer does not alter optimisation dynamics appreciably** at this scale (Figure 6).

4.2 Claims that are *not* supported

1. **“QAPINNs outperform classical PINNs.”** Not at matched parameters: 3 wins, 3 losses.
2. **“The quantum layer wins on the hard benchmark.”** It loses by 1.43×.
3. **“~20% parameter reduction” (Shah et al.).** Untested by this experiment — both models had *identical* parameter counts by construction. Testing that claim requires a *larger* classical network matched on *accuracy* instead.

5 Recommended actions

1. **Re-run CT.burgers_n3 with a different seed** (≈ 3 min). A valid number is required before the table can be published.
2. **Consider 2–3 seeds per configuration** for the six Tier-1 pairings. The Heat results swing between 0.27× and 1.74×, which is wide enough that single-seed conclusions are fragile. This is the single highest-value remaining experiment.
3. **Reframe the Phase-3 narrative** around *when* the quantum layer helps rather than *whether* it beats classical. The data answers the first question well and the second unfavourably.
4. **Test the Shah parameter-reduction claim properly**, if time permits, by training a *wider* classical PINN to match QAPINN *accuracy* and comparing sizes.

Summary. The classical twin experiment did what a control is supposed to do: it prevented an overclaim. The QAPINN’s advantage is real but narrow and problem-specific, the headline hard-benchmark run is beaten by its own matched baseline, and one run must be discarded. Reported honestly — with the frequency-content explanation that both this experiment and the Heat saturation support — this is a **stronger and more publishable** submission than an unqualified quantum-advantage claim would have been.